

Harvey J. Houlahan

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Machine learning engineer and geospatial researcher based in Byron Bay. Full-time at ArborMeta building the sovereign geospatial data infrastructure behind Australian carbon-credit policy — canopy and carbon models from LiDAR, advising federal and state government on accreditation pathways — while contracting independently for e-commerce and software clients on retainer. Advanced Computer Science at Monash (AI/ML and Advanced Algorithms concentrations, High Distinction). Off the clock: GPU-native ML systems, live at hjhportfolio.com — a WebGPU earth engine with a trained neural-operator surrogate, and a 40M-parameter LLM pretraining study.

EXPERIENCE

ArborMeta

Machine Learning Engineer

Byron Bay, NSW

Feb 2026 – Current

- Architected geospatial web applications (React, MapLibre GL JS, PostGIS) visualising ecological survey and carbon-farming data across Australia — the live platform informing land management and carbon-credit quantification.
- Built canopy-height and above-ground-carbon models from paired LiDAR captures, quantifying growth between flights parcel by parcel — the numbers underneath the policy advice.
- Conducted primary research into carbon sequestration measurement and verification, producing analysis that consults federal and state government on accreditation pathways for Australian carbon credits — a pathway toward doctoral research in computational ecology.
- Engineered high-performance spatial pipelines — multi-LOD PMTiles generation, Fulcrum field-survey parsing, ecological transect analysis — bringing field data from four countries into one live interface.

Step One Clothing

Software Engineer — contract, monthly retainer (concurrent with full-time role)

Barangaroo, Sydney

Aug 2025 – Current

- Primary front-end contractor: ongoing feature development, platform maintenance and e-commerce engineering across the Shopify stack for a major Australian apparel brand.
- Developed an AI-powered search matching natural-language descriptions to product attributes, lifting discoverability and purchase intent across the catalogue.
- Engineered the premium mega-menu experience end-to-end with schema-driven Liquid controls, letting the merchandising team edit live content without engineering support.

FibreTrace — Mobile Passport iOS App

iOS Engineer — contract

Moree, NSW

Mar 2025 – Dec 2025

- Conceived, pitched and built end-to-end a native SwiftUI product-passport app for FibreTrace's world-first fibre-level verification: QR/NFC scanning, signed deep-linked journeys, offline-first caching, role-based UI (Consumer vs Brand), certificates with PDF export, and App Clip entry at <3s time-to-passport.
- Shipped on-device intelligence: local JWT/HMAC checks (Keychain / App Attest) to flag spoofed tags, anomaly detection on chain-of-custody, and ETA prediction.
- A TestFlight demo and 60-second pitch secured CEO buy-in and a partner-brand pilot integrated with the FibreTrace platform.

Friday Technologies — Apple Consultancy

AI/Backend Engineer — contract

South Melbourne

May 2025 – Jan 2026

- Designed and deployed backend systems and ML infrastructure for production client applications; integrated APIs and built data pipelines feeding intelligent features.
- Contributed to projects recognised by Apple, working with Core ML and visionOS.

Australian Energy Market Operator (AEMO)

Software Engineer Intern

Melbourne CBD

Casual

- Built Python analytics over large NEM/MMS market datasets (dispatch, settlements, PASA) with real-time dashboards used by operations for demand and renewables insight.
- Engineered high-throughput C parsers for market and telemetry files (streaming validation, mmap-based I/O) with Python bindings feeding ETL — materially cutting ingest and compute time.
- Delivered C# micro-services against AEMO's MMS schemas and upstream SQL stores, accelerating daily reporting and ST/MT PASA reliability tooling.

SELECTED TECHNICAL WORK

Catchment — a neural earth engine · hjhportfolio.com/catchment

Real-terrain hydrology, erosion and wildfire simulated in hand-written WebGPU/WGSL at 60 fps, with a convolutional neural operator trained on the solver and transcribed to raw GPU compute — physics teacher and neural student racing live, error field on screen. Driven by a natural-language terminal whose 26k-parameter intent model was trained in numpy and ships as 83 KB of JSON. No libraries, no runtime, no server.

LLM pretraining under fixed compute · hjhportfolio.com/pretraining

A ~40M-parameter transformer pretraining study: validation loss 1.75 → 1.18 (−33%) in seven epochs on a fixed compute budget — Muon, rotary embeddings, warmup–stable–decay, ReLU², value residuals — each change argued from first principles rather than swept.

Genesis — an artificial-life lab · hjhportfolio.com/genesis

Thousands of GPU particles with flocking, wind fields and cyclic predation; describe a lifeform in plain English and CLIP scoring plus CMA-ES evolution steer the swarm toward it — entirely in-browser.

EDUCATION

Monash University

B.S. in Advanced Computer Science

Melbourne, Australia

Feb 2023 – Nov 2025

- Concentrations: Artificial Intelligence / Machine Learning, Advanced Algorithms & Data Structures · GPA 3.6/4.0, High Distinction · Teaching Assistant, FIT2014 Algorithms and Data Structures.

James Cook University

M.B.B.S. Medicine, direct entry — deferred after six months: writing more code than medical notes 2022

Townsville, Australia

The Southport School

ATAR 99.4 · Senior Dux of Boarding · Proxime Accessit Dux, Cohort 2021

Southport, Australia

2016 – 2021

ACTIVITIES & LEADERSHIP

- Social Executive, University College (University of Melbourne) — voted executive of a 300+ member college; ran the social calendar and alumni sessions. 2023
- Residential Tutor & Rugby/Swimming Coach, Townsville Grammar School — pastoral care, tutoring and coaching for boarders, Years 7–12. 2022

SKILLS

Languages: Python, TypeScript/JavaScript, Swift (iOS), C/C++, C#, Java, SQL, WGSL, Rust

ML & Systems: PyTorch, numpy, FastAPI, PostGIS, React, MapLibre GL JS, WebGPU, Shopify/Liquid, Core ML, AWS, Git